HOPE INTERNATIONAL | INFORMATION SECURITY

# Hope-Armor SIEM

## Development Guide — Architecture & Technology Stack

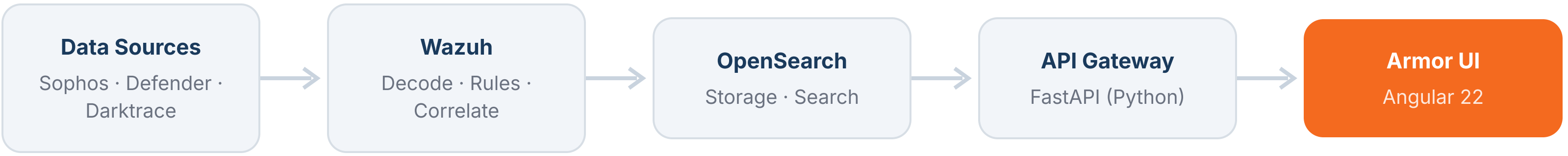
FRONTEND
**Angular 22 · Angular Material**

BACKEND
**Python 3.11 · FastAPI**

CORE DATA STORE
**Wazuh · OpenSearch**

## Architecture at a Glance

Hope-Armor is a single-pane-of-glass SIEM UI. Every integration below is normalized and correlated inside Wazuh, stored in OpenSearch, exposed through one Python API Gateway, and rendered by the Angular frontend — with Grafana sitting alongside for dashboarding on the same underlying data.



*Grafana reads the same OpenSearch indices in parallel and is embedded via the Visualizations tab (not shown above for clarity).*

## Frontend Stack — siem-dashboard/

| LIBRARY | VERSION | ROLE IN THIS CODEBASE |
|---|---|---|
| **Angular** | `^22.0.0` | Core framework — standalone components throughout, no NgModules. Built with the newer esbuild-based `@angular/build` and a Vite-powered dev server. |
| **Angular Material + CDK** | `^22.0.0` | Component library — dialogs, tables, tooltips, spinners, snackbars across every feature page. |
| **RxJS** | `~7.8.0` | Reactive streams for HTTP calls, debounced search, and notification polling. |
| **zone.js** | `~0.15.0` | Change-detection scheduling. See the engineering note at the end of this guide before relying on it. |
| **Chart.js + ng2-charts** | `^4.5.1 / ^10.0.0` | Trend charts on the Vulnerability Detection (Nessus) tab. |
| **Tailwind CSS** | `^3.4.19` | Utility classes alongside hand-written SCSS design tokens; `preflight` is disabled so its resets don't fight Angular Material. |
| **TypeScript** | `~6.0.0` | Language for the entire app. |
| **Karma + Jasmine** | `~6.4.0 / ~5.2.0` | Default Angular CLI test scaffold. |
| **PostCSS + Autoprefixer** | `^8.5.19 / ^10.5.4` | CSS build pipeline supporting Tailwind. |

**Cleanup candidate:** `@azure/msal-angular` and `@azure/msal-browser` are still listed in `package.json` but nothing imports them — auth was migrated to a lightweight custom Azure AD (Entra ID) implicit-flow client in `core/services/auth.service.ts`. Safe to remove from dependencies.

## Backend Stack — APIGateway/

| LIBRARY | VERSION | ROLE IN THIS CODEBASE |
|---|---|---|
| **Python** | `3.11` | Runtime — `python:3.11-slim` Docker base image. |
| **FastAPI** | `>=0.111.0` | Web framework for the entire API Gateway (`main.py` + routers). |
| **Uvicorn [standard]** | `>=0.29.0` | ASGI server (uvloop + httptools extras) — the container's actual entrypoint process. |
| **python-jose [cryptography]** | `>=3.3.0` | Verifies Azure AD–issued RS256 JWTs against signing keys cached from Azure AD's federation metadata (`auth.py`). |
| **PyJWT [crypto]** | `>=2.8.0` | Secondary JWT handling used alongside python-jose. |
| **httpx** | `>=0.27.0` | Async HTTP client — Grafana, Sophos, Defender/Graph, and most other outbound integration calls. |
| **requests** | `>=2.31.0` | Sync HTTP client used by a few older client classes (Sophos, Nessus, Jira, Wazuh Manager API) in `client.py`. |
| **opensearch-py** | `>=2.4.0` | Client for the Wazuh Indexer / OpenSearch cluster — every alert query goes through this. |
| **openpyxl** | `>=3.1.0` | Excel export for vulnerability reports. |
| **reportlab** | `>=4.2.0` | PDF generation for executive/technical vulnerability reports. |
| **python-multipart** | `>=0.0.9` | Required by FastAPI's `UploadFile`/form handling (vulnerability scan uploads). |
| **python-dotenv** | `>=1.0.0` | Loads `APIGateway/.env` (all integration credentials, never committed to git). |
| **tldextract** | `>=5.1.0` | Domain parsing for VirusTotal URL/domain lookups. |

## Infrastructure & External Systems

Not libraries in a package.json — but just as core to what this app actually is.

### Wazuh (Manager + Indexer)

Core SIEM engine — log decoding, detection rules, cross-source correlation. Indexer runs on OpenSearch.

### Grafana

Dashboard/visualization layer on the same OpenSearch data, provisioned via the API Gateway.

### nginx (Alpine)

Serves the built Angular app, terminates TLS, and reverse-proxies `/api/` to the FastAPI container.

### Docker Compose

Orchestrates the `backend` and `frontend` containers side by side.

**External integrations**

- Sophos Central
- Microsoft Defender / Entra ID
- Darktrace (via Wazuh)
- VirusTotal
- MISP (Wazuh-side)
- Jira
- Microsoft Teams (webhooks)
- Graylog (planned)

## Repository Layout

```
SIEM/
├── siem-dashboard/        # Angular 22 frontend
│   ├── src/app/features/  # one folder per tab (dashboard, alerts, nessus, docs, ...)
│   ├── src/app/core/      # services, interceptors, guards
│   ├── src/app/layout/    # shell, header, sidebar
│   ├── Dockerfile          # node:22-alpine build -> nginx:1.29-alpine serve
│   └── nginx.conf
├── APIGateway/             # FastAPI backend
│   ├── main.py             # route definitions
│   ├── integrations/       # ms_defender.py, virustotal.py, ...
│   ├── client.py           # Sophos / Nessus / Jira / Wazuh Manager clients
│   ├── routers/            # notifications, saved searches, alert status
│   ├── requirements.txt
│   └── Dockerfile          # python:3.11-slim
└── docker-compose.yml       # orchestrates backend + frontend
```

### ⚠ Known engineering note — change detection

In this deployment's environment, zone.js does not reliably auto-trigger Angular's change detection after async work completes (confirmed: `XMLHttpRequest.prototype.send` stays unpatched). The workaround already in place across every feature component is an explicit `ChangeDetectorRef.detectChanges()` call at the end of each HTTP subscribe callback that updates template-bound state. Any new component that fetches data must follow the same pattern, or its view will silently never update after the request completes.